Artificial Intelligence Lab

Dibyasundar Das

10 September 2026

LAB 07

- Write a Python program that plays Connect-3 on a 4 × 4 board perfectly, using
  - plain minimax search
  - minimax search with alpha–beta pruning
- Your aim is to measure how much of the game tree the pruning actually removes. The board is small enough to search all the way to the end of the game.
- **The game**
  1. The board has 4 columns and 4 rows, so 16 cells in all.
  2. Two players, X and O, move alternately. X moves first.
  3. A move is the choice of a column. The piece falls to the lowest empty cell of that column (gravity). A full column is not a legal move.
  4. A player wins immediately on making three of their own pieces in a line — horizontal, vertical, or either diagonal.
  5. If all 16 cells are filled and no player has a line, the game is a draw.

Output for minmax algorithm

```
---- Connect 3 Using MinMax Algorithm ----

User is X
AI is O

Here is the empty board:

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |

User Choice of Drop : 2

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
```

```
| - | - | - | - |
| - | X | - | - |

AI Thinking ... AI Choice : 3

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |
| - | X | O | - |

User Choice of Drop : 4

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |
| - | X | O | X |

AI Thinking ... AI Choice : 3

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | O | - |
| - | X | O | X |

User Choice of Drop : 2

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | X | O | - |
| - | X | O | X |

AI Thinking ... AI Choice : 3

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | O | - |
| - | X | O | - |
| - | X | O | X |
```

```
AI Won.... Number of Searches done in total : 219,059
```

Output for alpha-beta Pruning

```
---- Connect 3 Using Alpha Beta Pruning ----

User is X
AI is O

Here is the empty board:

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |

User Choice of Drop : 2

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |
| - | X | - | - |

AI Thinking ... AI Choice : 3

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |
| - | X | O | - |

User Choice of Drop : 4

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | - | - |
| - | X | O | X |

AI Thinking ... AI Choice : 3

| 1 | 2 | 3 | 4 |
```

```
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | - | O | - |
| - | X | O | X |

User Choice of Drop : 2

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | - | - |
| - | X | O | - |
| - | X | O | X |

AI Thinking ... AI Choice : 3

| 1 | 2 | 3 | 4 |
|:-:|:-:|:-:|:-:|
| - | - | - | - |
| - | - | O | - |
| - | X | O | - |
| - | X | O | X |

AI Won.... Number of Searches done in total : 771
```